



10242 - Understanding the origin of the [CII] deficit and/or PAH-Mid- infrared excess in galaxy M51b

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Jorge L. Pineda (PI)	Jet Propulsion Laboratory
Dr. Paul Goldsmith (CoI)	Jet Propulsion Laboratory
Dr. Lee Armus (CoI)	California Institute of Technology

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NIRCAM	NIRCam Imaging	(1) NGC-5195
	2	MIRI	MIRI Imaging	(3) NGC-5195-MIRI
	3	MIRI OFF	MIRI Imaging	(2) NGC-5195-OFF

ABSTRACT

We propose to use JWST near- and mid- infrared images of galaxy M51b to resolve (5-15pc resolution) and disentangle the distribution of PAH, hot dust, ionized gas, the stellar population, and AGN activity in the disk and nucleus of this galaxy. This galaxy is the closest example of an active galactic nucleus showing a low [CII]/FIR, [CII]/PAH and [CII]/MIR ratios relative to normal galaxies, the so called [CII] deficit, which is typical of distant ultra luminous infrared galaxies.

JWST observations of PAH and dust continuum emission will enable us to resolve and separate the different heating sources, which in turn can be used for an accurate comparison with lower resolution Herschel and SOFIA [CII] and [OI]. Such comparison will enable us to understand the thermal balance in this galaxy and the origin of the [CII] deficit. These observations have the potential to shed light on a long standing problem in the use of the [CII] and infrared emission to trace star formation of galaxies and therefore will be an important contribution to our understanding of the growth

OBSERVING DESCRIPTION

Observing Strategy: We propose to image line and continuum emission between 1.8m and 11.3m over the full extent of galaxy M51b with the NIRCAM and MIRI instruments on JWST in imaging mode (Fig. 2). The angular extent of M51b is smaller than the size of a single NIRCAM module and MIRI FULL sub-array and therefore the observations can be accomplished within a single pointing with each of these instruments. We will use the INTRAMODULEBOX10 and 4-Point-Sets dither patterns to ensure uniform intramodule coverage on both NIRCAM and MIRI observations, respectively.

Filter selection: For the short wavelength NIRCAM filters, we selected the F187N (P), F212N (H 2), F164N ([FeII]) narrow filters and the F200W wide filters for stellar distribution imaging and continuum subtraction of the P, H 2 , and [FeII] lines. For the long wavelength NIRCAM filters, we selected F335M (3.6m PAH feature) and F300M, F360M, and F444W for continuum subtraction. For MIRI we selected the F770W, F1130W, and F1800W filters to obtain images of the 7.7m and 11.3m PAH features and H 2 , and the F1000W, F1500W, F2100W filters for continuum subtraction, and to image the hot dust distribution in M51b. The F770W and F1130W filters are dominated by PAH emission and provide a good template for the spatial distribution of PAHs, and thus the heating sources.

Observing Time estimate: We estimated the intensities as a function of wavelength using the template spectrum from (22) which includes Spitzer IRS spectrum covering the full disk of M51b. We modeled the emission of M51b assuming a 2D-Gaussian distribution with a FWHM of 6.00 based on the Spitzer/IRAC 8m image of this galaxy. For NIRCAM, we use will the BRIGHT2 readout pattern with 2 groups per integration and 3 integration per exposure. With 4 dithers this leads to an exposure of 601 seconds per filter pair. These integration time results in SNR>20 in all NIRCAM filters. Given that M51b is contained within a single NIRCAM module, we will use module B only, to reduce excess data to be below the lower threshold. For MIRI we will use the FASTR1 readout pattern with 10 groups with two integrations each and one exposure per each of the four dithers (4-Point-

Sets pattern). This configuration results in 233 seconds of integration time per filter which will result in a $\text{SNR} > 20$ for all MIRI filters. An identical background MIRI observation will be performed in an adjacent emission-free region. The total observing time request is 5.5h.

Proposal 10242 - Targets - Understanding the origin of the [CII] deficit and/or PAH-Mid- infrared excess in galaxy M51b

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-5195	RA: 13 30 0.0960 (202.5004000d) Dec: +47 15 56.19 (47.26561d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[LINER galaxies] Extended=YES				
	(2)	NGC-5195-OFF	RA: 13 30 17.9189 (202.5746621d) Dec: +47 16 9.48 (47.26930d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				
	(3)	NGC-5195-MIRI	RA: 13 30 0.0960 (202.5004000d) Dec: +47 15 56.19 (47.26561d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: Category=Galaxy Description=[LINER galaxies] Extended=YES				

Proposal 10242 - Observation 1 - Understanding the origin of the [CII] deficit and/or PAH-Mid- infrared excess in galaxy M51b

Observation	Proposal 10242, Observation 1: NIRCAM										Wed May 13 20:00:33 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	NGC-5195	RA: 13 30 0.0960 (202.5004000d)			Epoch of Position: 2015.5					
			Dec: +47 15 56.19 (47.26561d)								
			Equinox: J2000								
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[LINER galaxies] Extended=YES									
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F187N	F335M	BRIGHT2	2	3	12	4	601.259	223509	
	2	F164N+F150W2	F300M	BRIGHT2	2	3	12	4	601.259	223509	
	3	F212N	F360M	BRIGHT2	2	3	12	4	601.259	223509	
	4	F200W	F444W	BRIGHT2	2	3	12	4	601.259	223509	
Special Requirements	Aperture PA Range 175.05262691 to 320.05262691 Degrees (V3 175.0 to 320.0)										

Proposal 10242 - Observation 2 - Understanding the origin of the [CII] deficit and/or PAH-Mid- infrared excess in galaxy M51b

Wed May 13 20:00:33 GMT 2026

Observation	Proposal 10242, Observation 2: MIRI										
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observations:[MIRI OFF (Obs 3)]										
Diagnostics	(MIRI (Obs 2)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(3)	NGC-5195-MIRI	RA: 13 30 0.0960 (202.5004000d)			Epoch of Position: 2015.5					
			Dec: +47 15 56.19 (47.26561d)								
			Equinox: J2000								
Template	Comments: Category=Galaxy Description=[LINER galaxies] Extended=YES										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	2	F1000W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	3	F1130W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	4	F1500W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	5	F1800W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	6	F2100W	FASTR1	10	2	1	Dither 1	4	8	233.103	223509
Special Requirements	Sequence Observations 3, 2 (reordered), Non-interruptible										

Proposal 10242 - Observation 3 - Understanding the origin of the [CII] deficit and/or PAH-Mid- infrared excess in galaxy M51b

Observation	Proposal 10242, Observation 3: MIRI OFF										Wed May 13 20:00:33 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observation For: [MIRI (Obs 2)]										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC-5195-OFF	RA: 13 30 17.9189 (202.5746621d)								
			Dec: +47 16 9.48 (47.26930d)								
			Equinox: J2000								
Template	Comments: Category=Calibration Description=[Telescope/sky background]										
	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	2	F1000W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	3	F1130W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	4	F1500W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	5	F1800W	FASTR1	20	1	1	Dither 1	4	4	222.003	223509
	6	F2100W	FASTR1	10	2	1	Dither 1	4	8	233.103	
Special Requirements	Sequence Observations 3, 2 (reordered), Non-interruptible										